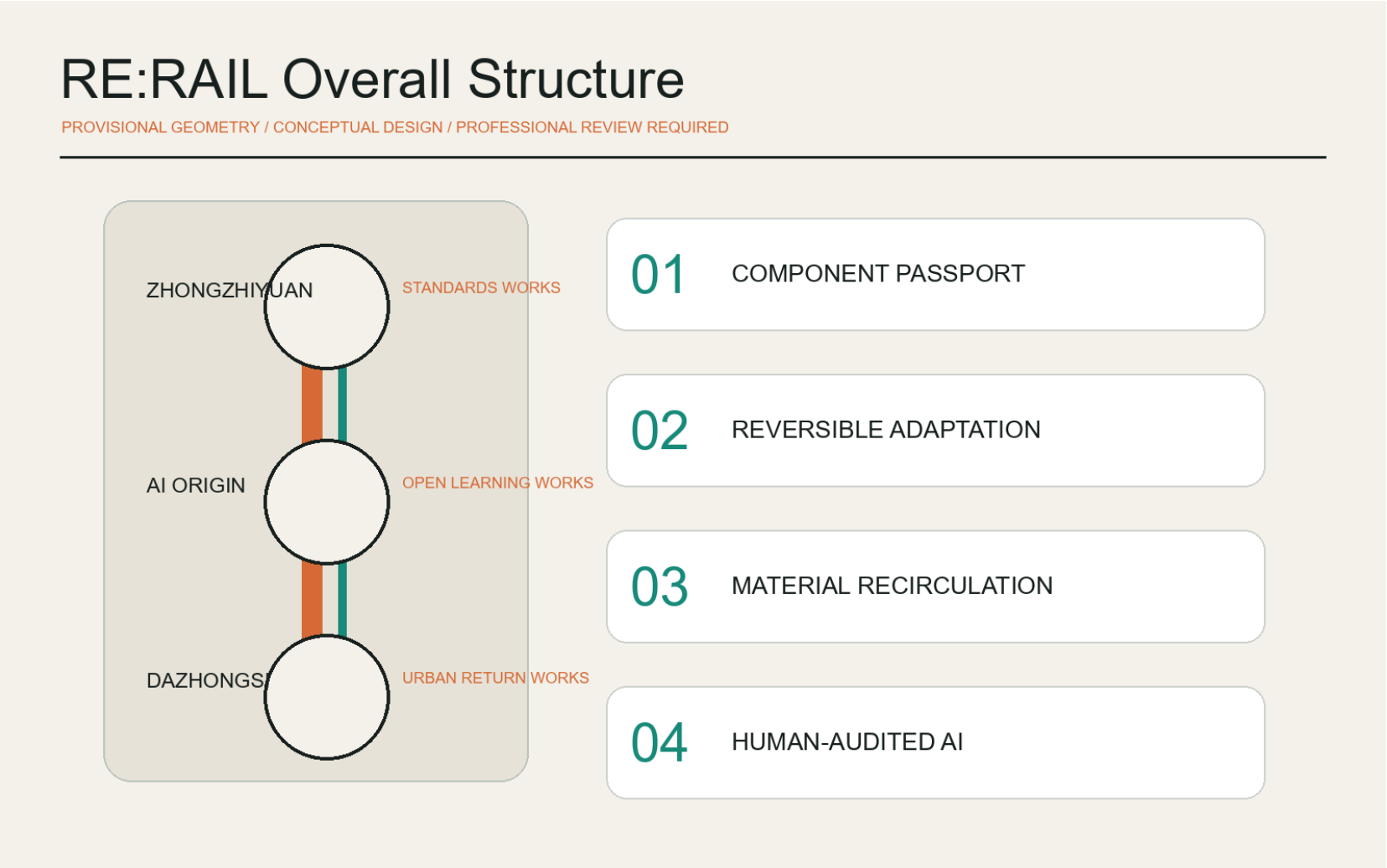


RE:RAIL

AI Circular Construction and Urban Renewal Commons

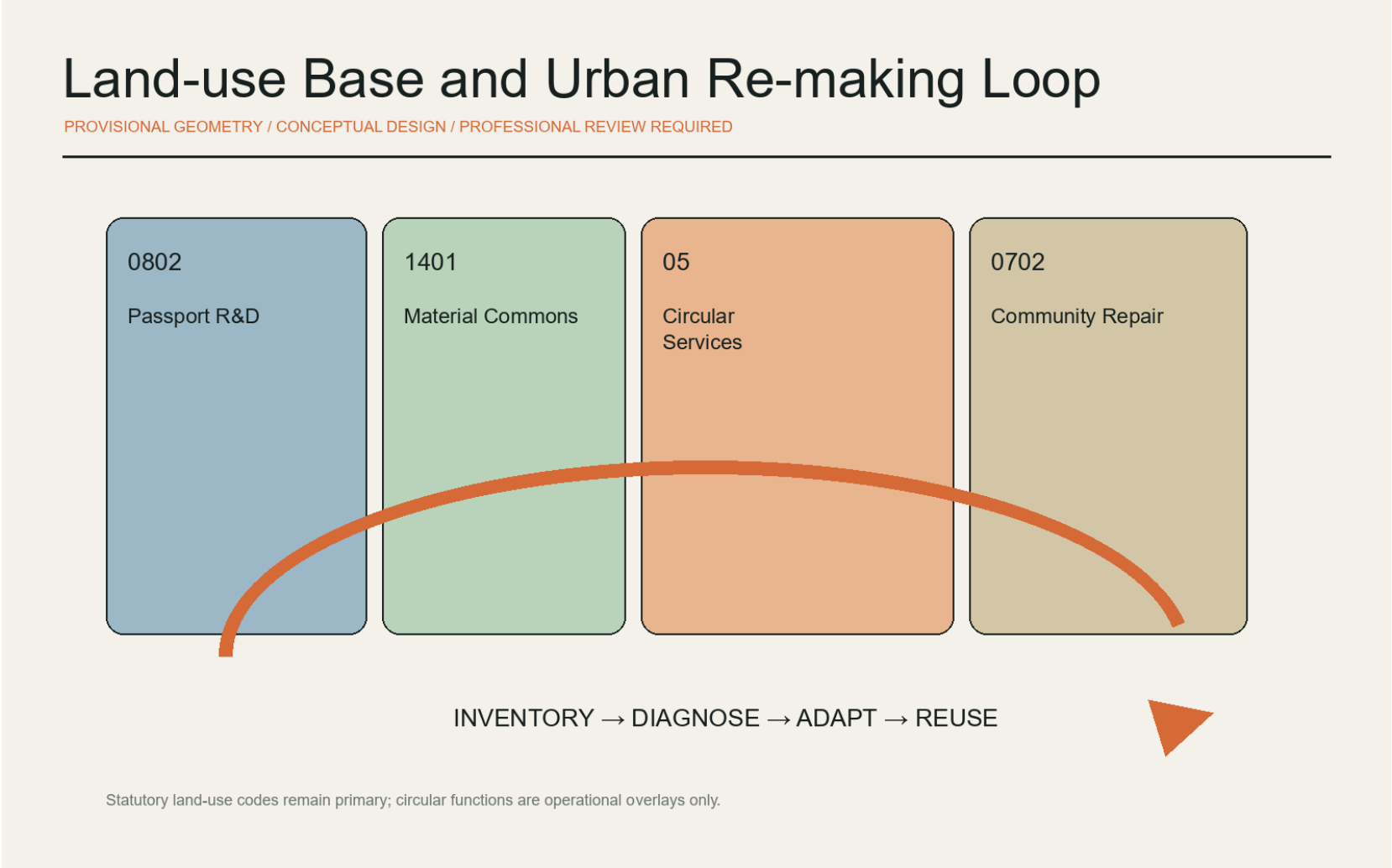
PROVISIONAL GEOMETRY / CONCEPTUAL DESIGN / PROFESSIONAL REVIEW REQUIRED

Design Basis and Source List



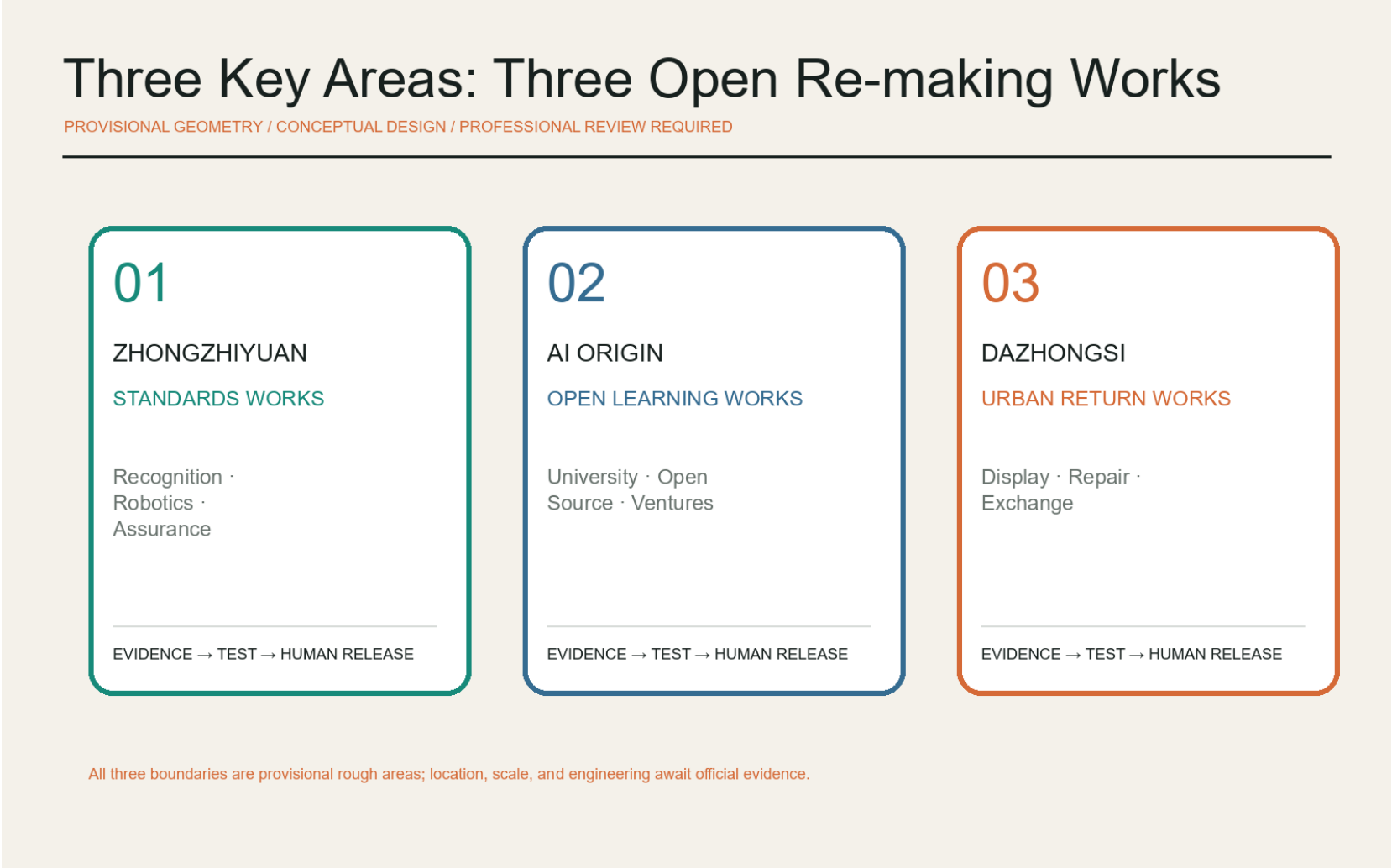
This proposal is grounded in the official prequalification announcement, the cleared Agent taskbook, the national urban-design measures, the control-plan procedure, and the national land-use classification guide. The announcement establishes the three working scales, three key areas, and the required urban-renewal agenda. The repository polygons are explicitly rough temporary geometry for generation and review; they are not official redlines and cannot support parcel ownership, precise area, engineering, or approval claims. [source:OFFICIAL-ANNOUNCEMENT] [standard:PROJECT-OFFICIAL-ANNOUNCEMENT] [depth:existing_conditions_diagnosis] The design proposition is RE:RAIL. “Re-making” does not mean wholesale demolition. It makes maintenance, adaptation, component replacement, and public-space repair into traceable urban knowledge. AI reads public or cleared records, generates alternatives, flags uncertainty, and prepares human review. Licensed professionals, right holders, public authorities, operators, and residents retain final judgment. Sources are separated into formal task evidence, ex

Coordinated Research Area: Industry and Future City Research



RE:RAIL treats the AI ecosystem as an urban-renewal operating system. Standards create a common language; building and component passports preserve asset memory; city models compare alternatives; test spaces validate them; repair networks create public value; and professional responsibility gates release decisions. Zhongzhiyuan hosts recognition, robotics, component testing, and AI assurance. The AI Origin Community connects university research, open source, and young ventures. Dazhongsi connects exhibition, service, exchange, and recirculation. [source:AGENT-TASKBOOK] [standard:PROJECT-AGENT-OPEN-CALL-TASKBOOK] Seven global precedents inform mechanisms, not local controls: STATION F demonstrates dense startup services; Mila connects research, industry, and entrepreneurship; Singapore AI Verify demonstrates open assurance; Amsterdam uses procurement and public space to support circularity; EU Level(s) provides a lifecycle language; Madaster demonstrates material passports; and EC3 demonstrates open environmental product data. Every precedent is recorded with a URL and usage limitatio

Detailed Design of Key Areas

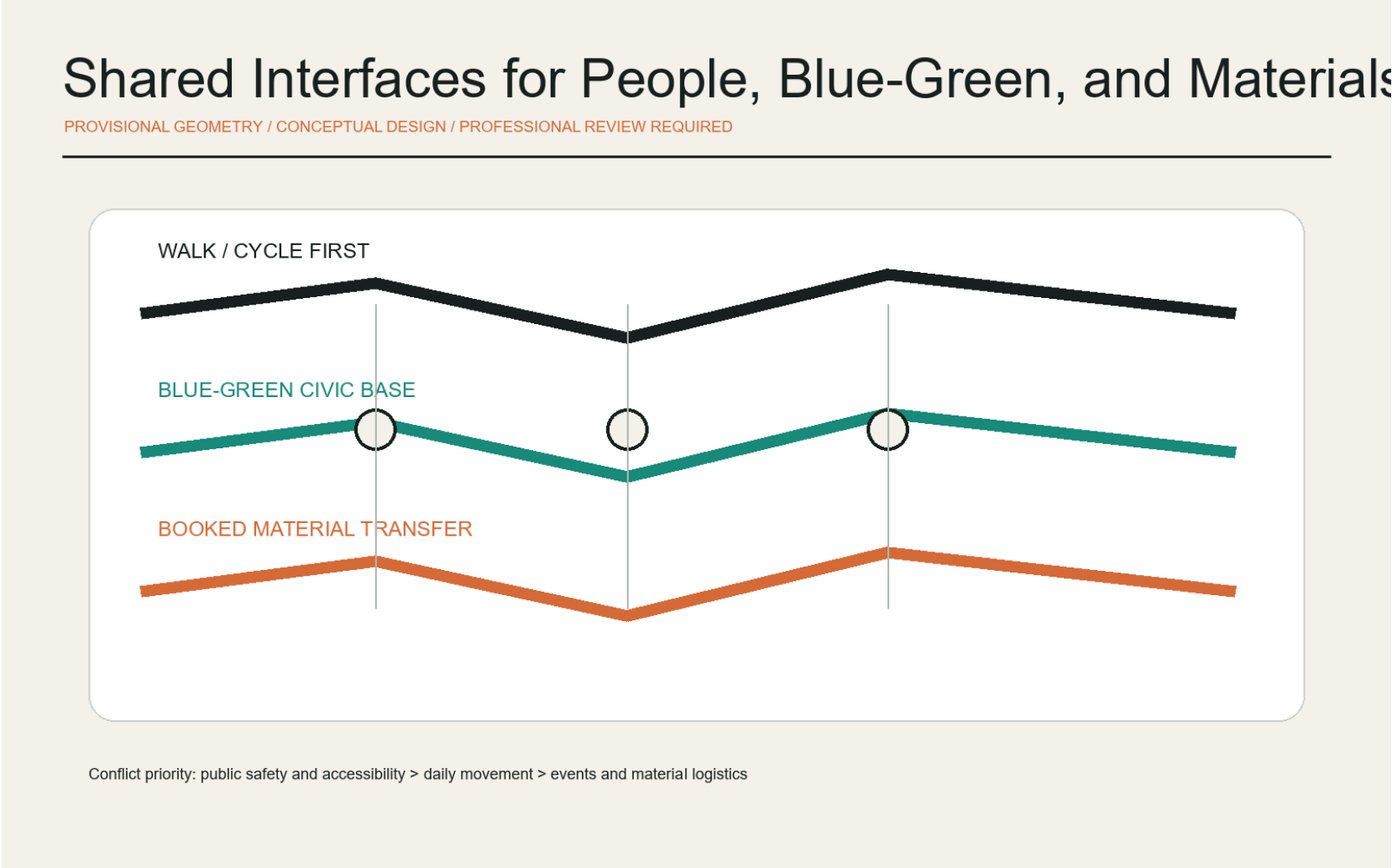


Zhongzhiyuan Standards Works combines a component-scanning and robotics test court, reversible joint mock-ups, an AI assurance room, and a Qinghe material garden. Tests begin in controlled or booked settings and require structural, fire, safety, and accessibility release before any public deployment. Failed tests enter a learning register. Exact locations remain pending official geometry and site evidence.

[data:geometry/key_areas.geojson#PROV-KEY-001] [depth:three_key_area_detailed_design] The AI Origin Open Learning Works links university research, open-source communities, young residents, and neighborhood repair. Its prototypes are an adaptation clinic, passport editing desk, student-craftsperson workshop, night study room, and small release space. No campus or privately controlled property is presumed available; initial work uses public interfaces and voluntarily cleared samples.

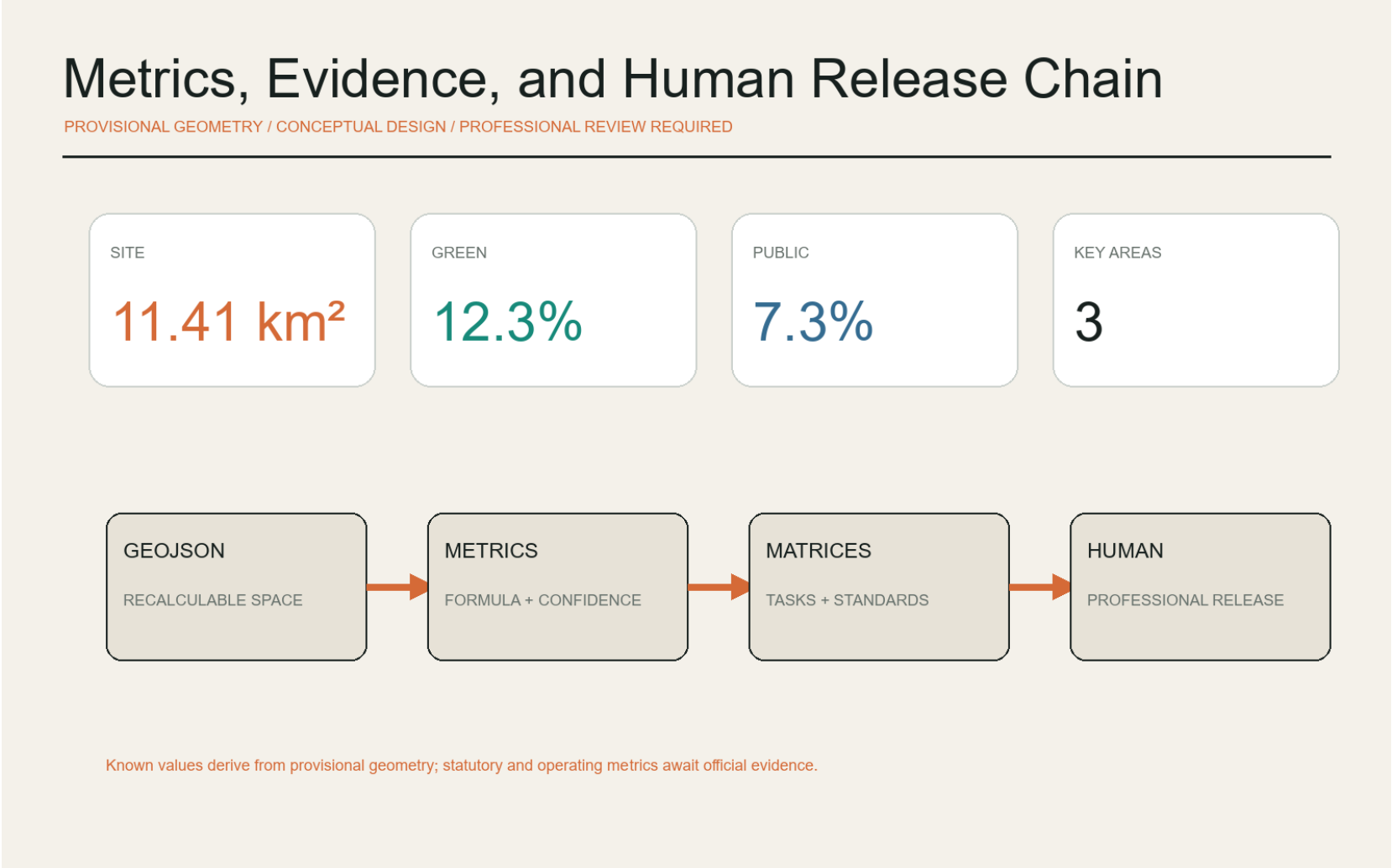
[data:geometry/key_areas.geojson#PROV-KEY-002] [source:CASE-MILA] Dazhongsi Urban Return Works overlays pedestrian continuity, commercial services, component display, repair booking, and citywide reuse m

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy



Land use retains codes 0802, 1401, 05, and 0702. Circular functions are secondary operational labels so that a design narrative cannot be mistaken for statutory land classification. The provisional partition covers the submitted boundary without overlap, supporting topology checks and comparative diagrams while awaiting official parcels and controls. [data:geometry/land_use.geojson#LU-002] [depth:land_use_layout] The building method uses five gates: R0 record and maintain, R1 repair, R2 adapt function, R3 reversible addition, and R4 consider replacement only through sufficient evidence and statutory procedure. Because the current building layer is conceptual, every object remains pending survey and ownership confirmation; no specific building is assigned a final action. [data:geometry/buildings.geojson#BLDG-001] [depth:retain_renovate_demolish] A component passport can record provenance, age, material, connection, inspection, repair history, disassembly conditions, reuse limits, accountable parties, and data rights. AI may extract fields from cleared BIM, inventories, and reports and

Blue-Green Network, Public Space, and Urban Character



The heritage park and the Qinghe and Xiaoyue River contexts form a low-impact civic base. Continuous shade, rain gardens, maintainable paving, and demountable exhibitions can support innovation without turning green space into a technology showroom. Any river, tree, drainage, or heritage action waits for official control lines and specialist review. Ratios are derived from provisional geometry and used only to compare the concept. [data:geometry/green_space.geojson#GREEN-001] [metric:green_ratio] [depth:blue_green_public_space] The public-space kit includes a passport marker, component shelf, repair table, reversible stage, staffed-service beacon, and issue-receipt wall. Three pilgrimage landmarks are the Open Joint Wall at Zhongzhiyuan, the Centennial Construction Archive at AI Origin, and the Urban Re-making Bell at Dazhongsi. They honor contributions, failures, and repairs rather than individuals or companies. [data:geometry/public_space.geojson#PUBLIC-001] [standard:MOHURD-URBAN-DESIGN-MEASURES] The identity uses two reconnecting rail lines: graphite represents the inherited city

12 AI+ scenarios / ■■■■■■

Building diagnosis, component passports, reversible joints, robotic disassembly, low-carbon procurement, repair translation, co-created accessibility audit, material library, shared tools, event-component return, Jing-Zhang construction-history guide, and public renewal retrospective.

Risks: official boundaries, controls, buildings, roads, utilities, heritage, ownership, and operating baselines remain missing. Every AI output requires human review, appeal, and exit.